SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH321
Engineering Mathematics II

Tutor-Marked Assignment

January 2026 Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 20 % of the final mark for MTH321 Engineering Mathematics II.

The cut-off date for this assignment is **12 March 2026, 2355hrs**.

Note to Students:

You are to include the following particulars in your submission: Course Code, Title of the TMA, SUSS PI No., Your Name, and Submission Date.

For example, ABC123_TMA01_Sally001_TanMeiMeiSally (omit D/O, S/O). Use underscore and not space.

Question 1

In the design of a new electric scooter, the cooling of the battery pack is to be achieved by natural airflow during the motion of the scooter as illustrated below. During maximum throttle, the battery pack generates heat of $P(t)$ Watts. You are tasked to evaluate the safety operating conditions of the battery during the design phase by answering the following questions.

Figure Q1

(a) By applying the conservation of energy to the battery pack during maximum throttle, formulate the mathematical model that governs its temperature variation and show that it results in the IVP:

$$T' + rT = \frac{P(t)}{mc} + rT_s, \ T(0) = T_s, \quad \text{where} \quad r = \frac{UA}{mc}.$$

The mass and specific heat capacity of the battery are m and c respectively. The area of the battery exposed to the airflow is A and the overall heat transfer coefficient is U. The scooter is not used for a long time before the maximum throttle is applied.

(6 marks)

(b) During maximum throttle, the battery pack generates heat according to the function

$$P(t) = Q\left(1 - e^{-t}\right), \quad t \geq 0,$$

where Q is a positive constant. Using the method of integrating factor, solve the IVP in Question 1(a) for the temperature function of the battery pack.

(14 marks)

(c) Given the following parameters, plot the temperature function obtained in Question 1(b) using a graph plotter and attach a screenshot of the graph. Explain your observations. If the safety limit for the battery pack to function properly is 60°C, is the method of cooling the battery suitable?

$U = 40\ W/m^2.K$, $A = 0.1\ m^2$, $m = 5\ kg$,
$c = 1000\ J/kg.K$, $Q = 180\ W$, $T_s = 35\ °C$.

(10 marks)

Question 2

A buffer stop for a train is shown below, where its purpose is to prevent the train from derailing due to accidentally overshooting the end of the tracks. To soften the impact of collision, a suspension system is being used that consists of the typical springs and dampers.

Figure Q2

(a) You may watch this video clip https://youtu.be/EVBdrPggXO4?si=sfGuLlHJihVZa4ey&t=29 from the 29th second to the 34th second to understand how the buffer stop works.

In the event of a collision at $t = 0$ and $x = 0$ with an impact velocity v_0, assume that the train of mass M is still being propelled by force $P(t)$. By drawing a free-body diagram of the train in collision with the suspension system, apply Newton's 2nd Law and show that the IVP for the train during collision can be derived as

$$Mx'' + cx' + kx = P(t), \; x(0) = 0, \; x'(0) = v_0,$$

where c is the overall damping coefficient and k is the overall spring stiffness.

(8 marks)

(b) If the train free-rolls to collide with the buffer stop at a velocity of 1 m/s with $P(t) = 0$, solve the IVP in Question 2(a) using the characteristic equation. The parameter values are M = 50,000 kg, c = 100,000 Ns/m and k = 100,000 N/m.

(7 marks)

(c) If the train is being braked with a force $P(t) = -50,000\, e^{-2t}$ Newtons while colliding with the buffer stop at a velocity of 2 m/s, determine the solutions of the IVP in Question 2(a) again using Laplace transform for M = 50,000 kg, c = 150,000 Ns/m and k = 100,000 N/m.

(8 marks)

(d) Plot your solutions in Questions 2(b) and 2(c) (in one graph) for $t \geq 0$ using a graph plotter and attach a screenshot of the graph. Explain your observations.

(8 marks)

(e) The buffer stop can also work using sliding friction in addition to the suspension system. You may watch this video clip https://youtu.be/EVBdrPggXO4?si=GO4EHJI_unhZlKTX&t=63 from the 63th second to 72th second to understand how it works.

Assuming that the buffer stop slides with a constant frictional force F upon collision, apply Newton's 2nd Law separately to the free-body diagrams of the train and the buffer stop (of mass m) to show that the IVP for the train-buffer system can be derived as

$$Mx'' + c\left(x' - d'\right) + k\left(x - d\right) = P\left(t\right),\ x\left(0\right) = 0,\ x'\left(0\right) = v_0,$$
$$md'' - c\left(x' - d'\right) - k\left(x - d\right) = -F,\ d\left(0\right) = 0,\ d'\left(0\right) = 0,$$

where $d(t)$ represents the displacement of the buffer stop from the moment of collision. Other terms have the same meanings as before.

(8 marks)

(f) Using Laplace transform, obtain a system of linear equations (SLE) from the system of ODEs in Question 2(e). Do not use any of the parameter values given earlier. Then, describe how you would solve the SLE for $x(t)$ and $d(t)$ (no actual solving required).

(6 marks)

Question 3

A function $f(t)$ is defined by

$$f(t) = \begin{cases} e^{-t}, & 0 \leq t < 1, \\ 1/e, & 1 \leq t < 2. \end{cases}$$

(a) Applying the odd and even expansions, compute the Fourier series of $f(t)$ for both expansions.

(18 marks)

(b) Plot $f(t)$ from both Fourier expansions obtained in Question 3(a) for 100 harmonics using a graph plotter and attach a screenshot of the graph. Explain which expansion is preferred.

(7 marks)

---- END OF ASSIGNMENT ----